# Statistical Foundations of Machine Learning

## Homework 2 Solutions

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1. A company rolls out a new feature to users. Each user independently receives the feature with probability 1/2. Suppose 5 users log in.

   (a) Find the probability that exactly 3 out of 5 users receive the feature. Write down both the formula and the computed value.

   (b) Find the probability that at least 4 users receive the feature. Write down both the formula and the computed value.

   (c) You are now given that not all 5 users receive the feature, i.e., at least one of them has not received the feature. Find the conditional probability that exactly 3 users receive it.

   **Solution.**

   Let $X$ denote the number of users who receive the feature. Since each user independently receives the feature with probability 1/2,

$$X \sim \text{Bin}\left(5, \frac{1}{2}\right).$$

   Therefore,

$$P(X = k) = \binom{5}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{5-k}.$$

   (a)

$$\begin{aligned} P(X = 3) &= \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(\frac{1}{2}\right)^2 \\ &= \binom{5}{3} \left(\frac{1}{2}\right)^5 \\ &= 10 \left(\frac{1}{32}\right) \\ &= \boxed{\frac{5}{16}}. \end{aligned}$$

   (b)

$$\begin{aligned} P(X \geq 4) &= P(X = 4) + P(X = 5) \\ &= \binom{5}{4} \left(\frac{1}{2}\right)^5 + \binom{5}{5} \left(\frac{1}{2}\right)^5 \\ &= \frac{5}{32} + \frac{1}{32} \\ &= \boxed{\frac{3}{16}}. \end{aligned}$$

   (c) The given information that not all users receive the feature means $X < 5$. Therefore,



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$$
\begin{aligned}
P(X = 3 \mid X < 5) &= \frac{P(X = 3)}{P(X < 5)} \\
&= \frac{10/32}{1 - P(X = 5)} \\
&= \frac{10/32}{1 - 1/32} \\
&= \frac{10/32}{31/32} \\
&= \boxed{\frac{10}{31}}.
\end{aligned}
$$

2. Teams A and B play a best-of-five series, with the first team to win three games winning the series. The series ends immediately when one team wins three games. Team A wins a game with probability $p$, independently of the results of the other games.

   Let $X_k$ be the indicator random variable for Team A winning the series in $k$ games.

   (a) For $X_k$, determine its probability mass function as a function of $p$.

   (b) Let $X$ be the indicator random variable that Team A wins the series. Determine the probability mass function of $X$.

   **Solution.**

   (a) Since Team A needs three wins to win the series, it is impossible for Team A to win the series in 1 or 2 games. Hence,

   $$X_1 = X_2 = 0 \quad .$$

   For $k = 3, 4, 5$, Team A wins the series in exactly $k$ games if it wins exactly two of the first $k - 1$ games and then wins the $k$-th game.
   Therefore,

   $$P(X_k = 1) = \binom{k-1}{2} p^2 (1-p)^{k-3} p.$$

   Hence,

   $$P(X_k = 1) = \binom{k-1}{2} p^3 (1-p)^{k-3}, \qquad k = 3, 4, 5.$$

   Since $X_k$ is an indicator random variable,

   $$P(X_k = 0) = 1 - P(X_k = 1).$$

   Thus,

   $$p_{X_k}(x) = \begin{cases} \binom{k-1}{2} p^3 (1-p)^{k-3}, & x = 1, \\ 1 - \binom{k-1}{2} p^3 (1-p)^{k-3}, & x = 0, \\ 0, & \text{otherwise.} \end{cases}$$

   In particular,

   $$P(X_3 = 1) = p^3,$$



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$$P(X_4 = 1) = 3p^3(1-p),$$

and

$$P(X_5 = 1) = 6p^3(1-p)^2.$$

(b) Team A can win the series in three, four, or five games. These events are mutually exclusive. Therefore,

$$\begin{aligned} P(X=1) &= p^3 + 3p^3(1-p) + 6p^3(1-p)^2 \\ &= p^3\left[1 + 3(1-p) + 6(1-p)^2\right] \\ &= 10p^3 - 15p^4 + 6p^5. \end{aligned}$$

Since $X$ is an indicator random variable,

$$P(X=0) = 1 - \left(10p^3 - 15p^4 + 6p^5\right).$$

Therefore,

$$p_X(x) = \begin{cases} 10p^3 - 15p^4 + 6p^5, & x = 1, \\ 1 - 10p^3 + 15p^4 - 6p^5, & x = 0, \\ 0, & \text{otherwise.} \end{cases}$$

3. A random variable $X$ is said to have the Laplace$(a, b)$ distribution if it has the PDF

$$f_X(x) = C\exp\left(-b|x-a|\right), \qquad x \in \mathbb{R},$$

where $b > 0$.

(a) Plot the function for $a = 1$ and $b = 2$.

(b) Determine $C$ in terms of $a$ and $b$ so that $f_X$ is a valid density function.

(c) Find the CDF of $X$.

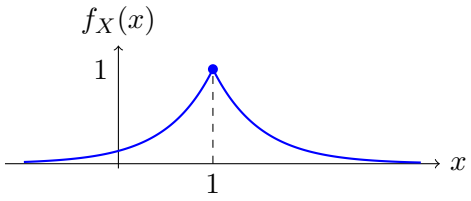
**Solution.**

(a) For $a = 1$ and $b = 2$,

$$f_X(x) = Ce^{-2|x-1|}.$$

From part (b), $C = b/2 = 1$. Therefore,

$$f_X(x) = e^{-2|x-1|} = \begin{cases} e^{2(x-1)}, & x < 1, \\ e^{-2(x-1)}, & x \geq 1. \end{cases}$$






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(b) For $f_X$ to be a valid PDF,

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

Splitting the integral at $x = a$,

$$\begin{aligned} 1 &= C\int_{-\infty}^{a} e^{b(x-a)}\,dx + C\int_{a}^{\infty} e^{-b(x-a)}\,dx \\ &= C\left(\frac{1}{b} + \frac{1}{b}\right) \\ &= \frac{2C}{b}. \end{aligned}$$

Hence,

$$\boxed{C = \frac{b}{2}}.$$

Therefore,

$$f_X(x) = \frac{b}{2}e^{-b|x-a|}.$$

(c) For $x < a$,

$$\begin{aligned} F_X(x) &= \int_{-\infty}^{x} \frac{b}{2}e^{b(t-a)}\,dt \\ &= \frac{1}{2}e^{b(x-a)}. \end{aligned}$$

For $x \geq a$,

$$\begin{aligned} 1 - F_X(x) &= \int_{x}^{\infty} \frac{b}{2}e^{-b(t-a)}\,dt \\ &= \frac{1}{2}e^{-b(x-a)}. \end{aligned}$$

Therefore,

$$F_X(x) = \begin{cases} \dfrac{1}{2}e^{b(x-a)}, & x < a, \\ 1 - \dfrac{1}{2}e^{-b(x-a)}, & x \geq a. \end{cases}$$

4. You are stuck in a cave with three tunnels in front of you. The first tunnel takes you out of the cave in 3 hours. The second tunnel brings you back to the starting position in 5 hours. The third tunnel also brings you back, but in 7 hours. The entrances to the tunnels are identical. What is the expected time for you to get out of the cave?

**Solution.**

Let $T$ be the total time required to leave the cave, and let

$$m = \mathbb{E}[T].$$

Let $A_i$ denote the event that tunnel $i$ is selected, for $i = 1, 2, 3$. Since the entrances are identical,

$$P(A_1) = P(A_2) = P(A_3) = \frac{1}{3}.$$



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The events $A_1, A_2, A_3$ form a partition of all possible first choices. Therefore, by the law of total expectation,

$$\mathbb{E}[T] = \sum_{i=1}^{3} P(A_i)\mathbb{E}[T \mid A_i].$$

If the first tunnel is selected, the required time is 3 hours, so

$$\mathbb{E}[T \mid A_1] = 3.$$

If the second tunnel is selected, 5 hours are spent and we return to the initial position. The expected additional time is again $m$. Hence,

$$\mathbb{E}[T \mid A_2] = 5 + m.$$

Similarly,

$$\mathbb{E}[T \mid A_3] = 7 + m.$$

Thus,

$$m = \frac{1}{3}(3) + \frac{1}{3}(5 + m) + \frac{1}{3}(7 + m).$$

Therefore,

$$\begin{aligned} m &= 5 + \frac{2m}{3}, \\ m - \frac{2m}{3} &= 5, \\ \frac{m}{3} &= 5. \end{aligned}$$

Hence,

$$\boxed{\mathbb{E}[T] = 15 \text{ hours}}.$$



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